

Section C

14) $a = b + c$ $d = b + c$ $b = a - d$ $e = b + c$

Expressions involved:-

• $E1: b + c$ • $E2: a - d$

Line	Statement	GEN	KILL
1	$a = b + c$	$\{b + c\}$	$\emptyset$
2	$d = b + c$	$\{b + c\}$	$\emptyset$
3	$b = a - d$	$\{a - d\}$	$\{b + c\}$
4	$e = b + c$	$\{b + c\}$	$\emptyset$

Available expressions use forward analysis:-

$$OUT[n] = GEN[n] \cup (IN[n] - KILL[n])$$

$$IN[n] = \bigcap OUT[pred]$$

• Line 1:-

$$IN = \emptyset$$

$$OUT = \{b + c\}$$

• Line 2:-

$$IN = \{b + c\}$$

$$OUT = \{b + c\}$$

• Line 3:-

$$IN = \{b + c\}$$

$$OUT = \{a - d\} \text{ (} b + c \text{ killed)}$$